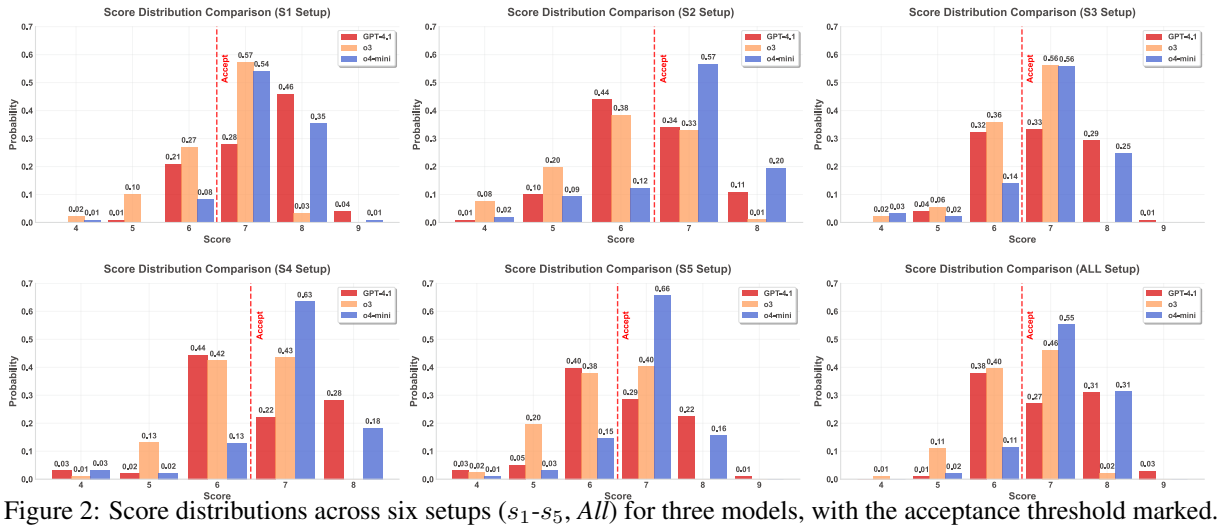

Figure 2: Score distributions across six setups (s_1-s_5, *All*) for three models, with the acceptance threshold marked. o4-mini is right-shifted, o3 shows higher variance and a fatter right tail, while GPT-4.1 is more conservative.

single strategies, s_1 provides the strongest acceptance with only moderate detection pressure (ICR@M 39.5%), whereas others (e.g., s_3-s_5) are somewhat weaker but also less detectable (ICR@M $\approx$ 30–34%). Across models, `o3` is the most flag-happy (consistently higher ICR-m), while `GPT-4.1` rarely flags concerns (mostly 2–8%), reinforcing that current review models have limited and uneven detection capability.

Score distributions. Figure 2 plots score histograms for three models across six setups (s_1-s_5, *All*) with the acceptance threshold marked. Overall, `o4-mini` is right-shifted—consistently placing more mass at ≥ 7—which aligns with its higher acceptance tendency. `o3` shows larger variance and a fatter right tail (notably in s_1 and *All*), producing many near-threshold and high scores; `GPT-4.1` is comparatively conservative, clustering around 6–7 with a thinner tail at 8+. Among strategies, s_1 yields the strongest rightward shift for all models, while s_2/s_4 are milder. The ALL setup increases polarization (more mass both just below and above the threshold), explaining why it sustains high acceptance yet is easier for detectors to flag.

Concern–Acceptance Conflict. Conditioned on a model posting an integrity concern, we report the share that still receives an acceptance-level score in Table 2. Conflict is widespread: `o4-mini` is most contradictory (s_1/s_3/*All*: 100%; s_2/s_4/s_5: 50–80%), `GPT-4.1` is mixed (0% in s_4 but 33–75% elsewhere), and `o3` is moderate (26–52%). s_3 (statistical theater) induces the largest cross-model conflict, and *All* further amplifies it for `o4-mini` (100%). These observations indicate even agents voice concerns, yet keep acceptance-high scores, and integrity signals are not well-coupled to review.

Table 3: ACPT and ICR for the baseline review agent vs. ReD. ReD lifts concerns but raises ACPTs.

	Baseline	ReD
ACPT-τ_{rate}	28.0%	44.0%
ACPT-$\tau_{0.5}$	37.0%	58.0%
ICR-o3	50.6%	84.0%
ICR-o4mini	12.4%	11.0%
ICR-GPT4.1	4.5%	0.0%
ICR@M	57.3%	86.0%

5 Mitigation

We study two complementary interventions that make the review pipeline explicitly integrity-aware. The first augments the standard review workflow with an integrity check; the second replaces reviewing with a lightweight triage detector.

5.1 Setup

Research Questions. RQ1 (Detection-Aware Reviewing). How can we enhance detection awareness *within* the act of reviewing—i.e., can adding an explicit integrity-detection subtask (risk classification with cited evidence) improve sensitivity to integrity risks *while* preserving or improving core review quality (helpfulness and recommendation accuracy)?

RQ2 (Triage without Full Reviews). When generating a full textual review is not the priority, can a detection-only agent reliably flag papers with potential integrity concerns at a low false-alarm rate

Table 4: Evaluation results of all detectors. Across various setups, detection offers only slight gains over random. ReD is more conservative, while DetOnly is recall-oriented with higher FPR. o3 shows a positive bias, whereas GPT-4.1 tends toward negative.

	o3				o4-mini				GPT-4.1			
Method	TPR	FPR	Acc	F1	TPR	FPR	Acc	F1	TPR	FPR	Acc	F1
Random Guess	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%	50.0%
ReD	81.6%	44.9%	67.0%	72.1%	0.0%	8.0%	46.0%	0.0%	0.0%	0.0%	50.0%	0.0%
DetOnly	98.0%	84.0%	57.0%	69.5%	64.0%	74.0%	45.0%	53.8%	24.0%	12.0%	56.0%	35.3%

and route low-risk papers to the standard pipeline?

Mitigation Strategies. (i) Review-with-Detection (ReD) We modify the review prompt to bundle a conventional rubric-based review with a mandatory detection of AI-generated content. For each document, a single model $m \in M$ outputs: (a) a standard review, (b) a binary label $y_m \in \{\text{NON-FABRICATED}, \text{AI-FABRICATED}\}$, and (c) *evidence snippets* (quoted lines) to support its finding. **(ii) Detection-Only (DetOnly)** We remove the reviewing rubric and instruct the agent to perform *only* the detection of AI-generated content. In this mode, the model's sole task is to return a binary label $y_m \in \{\text{NON-FABRICATED}, \text{AI-FABRICATED}\}$ and evidence, which serves as the direct and final triage action, without any secondary assessment.

Dataset. We construct a controlled set of 100 papers using the stratified sampling algorithm in Section 3.4: 50 are sampled from real ICLR 2025 submissions, and 50 are sampled from the generated corpus produced in Section 4.

5.2 Evaluation Analysis

To RQ1: Review Outcomes. We compare *ReD* against the initial review agent baseline (same models, same rubric, no integrity subtask). We report the review outcomes using the same metrics as in Section 4. As shown in Table 3, *ReD* increases integrity warnings overall (ICR@M $57.3\% \rightarrow 86.0\%$), driven largely by `o3` ($50.6\% \rightarrow 84.0\%$), while `o4-mini` is roughly unchanged and `GPT-4.1` collapses to 0%. Paradoxically, acceptance also rises substantially ($\text{ACPT}_{\tau_{\text{rate}}}$ $28.0\% \rightarrow 44.0\%$, $\text{ACPT}_{\tau_{0.5}}$ $37.0\% \rightarrow 58.0\%$). Thus, adding a detection subtask improves stated awareness but does not translate into stricter recommendations—if anything, it coexists with more accepts. This suggests the integrity signal is weakly coupled to scoring; practical deployments should gate or weight recommendations by risk rather than merely requesting detection within the review.

To RQ2: Detection Performance. We set three detectors on our new dataset: *Random Guess* baseline, the *ReD* integrity component, and *DetOnly*. The results are presented in Table 4. Overall, detection helps but just slightly: across models, accuracy is near the 50% random baseline, with a clear lift only on `o3` (ReD 67% vs. random 50%; DetOnly 57%). Comparing *ReD* and *DetOnly*, the latter is recall-seeking (higher TPR) but far noisier (much higher FPR), whereas ReD is more conservative and, on some bases, collapses (e.g., `GPT-4.1` shows 0% TPR for ReD). Model behavior also differs: `o3` tends to judge *positive* (high flag rate; e.g., DetOnly FPR 84%), while `GPT-4.1` tends to judge *negative* (low TPR/FPR), yielding a small accuracy gain for DetOnly (56%) over random.

6 Conclusion and Discussion

Our findings expose a critical vulnerability: LLM review systems can be systematically deceived by presentation manipulation. Fabricated papers achieve high acceptance rates, with reviewers frequently exhibiting concern-acceptance conflicts—flagging integrity issues yet still recommending acceptance. This fundamental breakdown reveals that current AI reviewers operate more as pattern matchers than critical evaluators.

Our mitigation attempts show the inadequacy of current defenses. Detection accuracy barely exceeds random chance, and paradoxically, adding explicit integrity checks sometimes increases acceptance rates. Simply asking LLM reviewers to "be more careful" is insufficient.

The scientific community faces an urgent choice. Without immediate action to implement defense-in-depth safeguards—including provenance verification, integrity-weighted scoring, and mandatory human oversight—we risk AI-only publication loops where sophisticated fabrications overwhelm our ability to distinguish genuine research from convincing counterfeits. The integrity of scientific knowledge itself is at stake.

Limitations

Scope. Our research focuses on presentation manipulation without executable code or real data generation, deliberately excluding prompt injection, forged credentials, and agent collusion to isolate this specific attack vector. Our scope is orthogonal to AI4Science misuse study (He et al., 2023; Jiang et al., 2025), which evaluate risks from scientific-knowledge misuse rather than reviewer-pipeline integrity. We evaluate three frontier LLMs with a standard rubric protocol; while results may vary across model families and augmented review systems, we expect similar failure modes given the fundamental pattern-matching vulnerabilities we identify. Real adversaries may employ hybrid strategies, though our approach already demonstrates systematic weaknesses.

Generalization. Our calibration uses ICLR 2025 data from AI/ML conference reviews. While acceptance rates and norms vary across disciplines and venues, our core finding—that presentation manipulation can deceive LLM reviewers—likely generalizes given the underlying pattern-matching limitations we identify. Adversarial adaptation remains an open challenge requiring ongoing research.

Evaluation Setup. We use `GPT-5` to classify integrity concerns in reviewer feedback and deliberately exclude human oversight to isolate LLM capabilities under adversarial pressure. This represents a controlled worst-case scenario; real workflows may include multiple human safeguards to mitigate potential failures. Our results provide critical stress-testing for systems increasingly relying on AI assistance.

Ethical Considerations

Research Intent and Dual-Use Risks. This work aims to strengthen scientific integrity by exposing vulnerabilities before malicious actors exploit them. We acknowledge dual-use concerns and mitigate through: keeping strategy descriptions abstract, emphasizing detection methods, coordinating responsible disclosure, and prioritizing defensive applications. We argue that transparent security research is preferable to covert vulnerability discovery.

Potential Harms and Misuse. *(i) Adversarial Guidance.* Malicious authors could exploit our strategies to improve fabrications. We mitigate by omitting prompt engineering details and withholding the complete generation codebase. *(ii) Automation Overconfidence.* Our modest improvements should not justify reduced human oversight. Detection accuracy barely exceeds chance, and current LLMs are not ready for autonomous review. *(iii) Reputation Harm.* Over-sensitive detectors may unfairly flag legitimate work with strong results, non-native writing, or novel claims. Deployment requires human arbitration and author appeal mechanisms.

Artifact Release. We will partially release our artifact due to ethical concerns. **Public release** includes: the evaluation framework, curated synthetic papers/reviews, detector models, and analysis scripts. **Restricted access** (authorized users upon request only): the complete paper generation agent with prompts, specific exploits, and large-scale fabrication scripts. All framework components require a responsible AI license with declaration of intended use and agreement not to fabricate academic content for distribution.

Deployment Recommendations. For venues considering AI-assisted review: (i) *Mandatory disclosure* of AI usage to authors and reviewers; (ii) *Score-flag coupling*—papers flagged with integrity concerns cannot receive acceptance without senior reviewer override; (iii) *Audit trails* logging all model inputs, outputs, and integrity evidence; (iv) *Human oversight* for all flagged submissions. **For researchers using AI disovery systems:** Authors remain fully responsible for verifying that AI-generated content accurately reflects their actual experiments, implementations, and results. Fabricated claims, whether intentional or due to AI hallucination, constitute scientific misconduct regardless of the generation method.

Broader Impacts. AI-only publication loops threaten scientific epistemology. If fabrications become indistinguishable from genuine work, the foundation of scientific knowledge risks collapse. The path forward requires defense-in-depth across multiple layers: *technical* (provenance verification, artifact validation), *procedural* (integrity-aware scoring, human oversight), *community* (post-publication review, whistleblower system), and *cultural* (education on AI limitations, ethical guidelines). We view this work as an *early warning* system to catalyze robust defenses before these failure modes manifest at scale. Our findings demon-